Industry-Level Design Document

Conversational ERP System with Policy-Aware & Auditable DBMS

1. Executive Summary

This document presents the design of an industry-grade Enterprise Resource Planning (ERP) system augmented with a conversational AI interface. The system is fundamentally a traditional ERP implementing full CRUD functionality through a web-based UI, while also supporting natural-language interaction via an AI/MCP server. The design emphasizes database integrity, scalability, security, and auditability, aligning with real-world enterprise standards.

2. System Goals & Non-Goals

Goals:

- Build a modular ERP with role-based access control
- Enforce business rules at the database level
- Maintain full audit trails and historical data
- Support dual interaction modes: Manual UI & Conversational AI
- Ensure ACID-compliant transactions

Non-Goals:

- Direct execution of AI-generated SQL without validation
- Replacing traditional ERP workflows with AI-only access

3. ERP Functional Modules

The ERP system is divided into independent yet integrated modules:

• User & Role Management
• Human Resources Management
• Inventory Management
• Order & Transaction Management
• Reporting & Analytics
• Audit & Compliance Module

4. High-Level Architecture

The system follows a layered and service-oriented architecture:

Frontend Layer: React/Next.js dashboards and forms
API Layer: FastAPI services handling business logic
AI Layer: MCP Server translating natural language to intent
Database Layer: PostgreSQL with constraints, triggers, and logs

5. Dual Interaction Model

The ERP supports two parallel interaction mechanisms operating on the same database:

Manual Mode: Traditional form-based CRUD via web UI
Conversational Mode: Natural language commands processed by AI
Both modes pass through the same validation, policy, and transaction layers

6. Database Design Philosophy

The database is designed in Third Normal Form (3NF) with strict enforcement of referential integrity. Primary keys, foreign keys, unique constraints, and indexes are used to ensure consistency and performance.

7. Audit, Temporal & Compliance Design

All data mutations are tracked using database triggers. Instead of hard deletes, records are soft-deleted with validity timestamps. Audit tables store previous and new values, operation type, user identity, and timestamps, enabling regulatory compliance and rollback analysis.

8. Policy-Aware CRUD Enforcement

Business rules are enforced primarily at the database layer using CHECK constraints, stored procedures, and triggers. The API layer only invokes approved procedures, ensuring that no rule can be bypassed by frontend or AI interactions.

9. Conversational AI & MCP Server Design

The MCP server acts as an intent-processing layer rather than a direct SQL generator. User input is classified into intents (CREATE, READ, UPDATE, DELETE), mapped to predefined backend functions, validated against RBAC policies, and executed within transactions.

10. Security Architecture

Security is enforced across all layers:
- JWT-based authentication
- Role-Based Access Control (RBAC)
- Parameterized queries and ORM usage
- AI output sanitization and approval layer
- Encrypted credentials and secrets management

11. Scalability & Reliability Considerations

The system is horizontally scalable at the API and AI layers. Database scalability is achieved through indexing, query optimization, and read replicas. All critical operations are wrapped in transactions to ensure reliability under concurrent access.

12. Technology Stack

Frontend: React / Next.js
Backend: Python (FastAPI)
Database: PostgreSQL
ORM: SQLAlchemy
AI Layer: MCP Server + LLM
Deployment: Docker (optional), Cloud-ready

13. Conclusion

This design represents an industry-aligned ERP architecture with a strong DBMS foundation. By combining traditional enterprise workflows with a controlled conversational interface, the system achieves usability without compromising data integrity, security, or compliance.